An Introduction to Ergodicity Economics: Markets (I)

Emilie Rosenlund Soysal

2026-03-26

Markets

6.5.3 Expected wealth and expected utility 86
6.6 From ergodicity transformation to wealth process and back 91
6.6.1 Itô setup 91
6.6.2 From ergodicity transformation to wealth process 92
6.6.3 From wealth process to ergodicity transformation . 95

7 Decisions in the real world 99
7.1 The St. Petersburg paradox 99
7.1.1 The St. Petersburg lottery 100
7.1.2 Expected-value theory – a paradox 101
7.1.3 Expected-utility theory – a resolution 101
7.1.4 Ergodicity economics – enter dynamics 102
7.2 The Copenhagen experiment 108
7.3 Insurance 111
7.4 A shipping insurance contract 114
7.4.1 Ergodicity economics solution 117
7.4.2 The classical solution of the insurance puzzle . . . 119

III Populations 121

8 People 123
8.1 Each one alone 123
8.1.1 Lognormal distribution 124
8.1.2 Two growth rates 125
8.1.3 Measuring inequality 129
8.1.4 Wealth condensation 132
8.1.5 Rescaled wealth 133
8.1.6 Power-law resemblance 134
8.2 Finite populations 136
8.2.1 How long is a long time? 136

9 Interactions 141
9.1 Cooperation 141
9.1.1 Farmers' fable 142
9.1.2 Model 143
9.2 Reallocation 150
9.2.1 The ergodic hypothesis in economics 151
9.2.2 Reallocating geometric Brownian motion 152
9.2.3 Positive reallocation 153
9.2.4 Negative reallocation 156
9.2.5 Fitting United States (US) wealth data 156

10 Markets 161
10.1 The portfolio selection problem 162

10.1.1 Riskless and risky assets 162
10.1.2 Leverage 163
10.1.3 Classical portfolio theory 166
10.1.4 Growth rate maximisation 170
10.2 Leverage efficiency 173
10.2.1 Relaxing the model 173
10.2.2 Efficiency 174
10.2.3 Stability 175
10.2.4 Prediction accuracy 177
10.3 Applications of leverage efficiency 180
10.3.1 A theory of noise 181
10.3.2 Solution of the equity premium puzzle 181
10.3.3 Central-bank interest rates 183
10.3.4 Fraud detection 184
10.4 Discussion 186

Acronyms 187

List of Symbols 189

Index 197

References 201

Markets

Economic agents can maximise their wealth over time by maximising the time-average growth rate of the wealth.

- What does that mean for the portfolio choice?
  - Optimal leverage
- What does that mean for the way markets organise?
  - Leverage efficiency

Geometric Brownian Motion

Geometric Brownian Motion (GBM) is the workhorse model of financial assets:

$$dx = x(\mu\, dt + \sigma\, dW)$$

with drift, μ, and volatility, σ.

Multiplicative dynamics:

Ensemble average growth rate: $g_{\langle\rangle} = \mu$

Time average growth rate: $\bar{g} = \mu - \frac{\sigma^2}{2}$

Portfolio choice

Assume there are only two assets available

- Risk-free asset:

$$dx_r = x_r \, \mu_r \, dt$$

- Risky asset (stock):

$$dx_s = x_s \left(\mu_s \, dt + \sigma_s \, dW \right)$$

Portfolio problem: Allocate wealth to each of the two assets. Leverage, ℓ, is the share of wealth allocated to the risky asset.

Optimal leverage

Wealth dynamic:

$$dx = x((\mu_r + \ell\,\mu_e)\,dt + \ell\,\sigma_s\,dW\,)$$

where μ_e is the excess return, $\mu_s - \mu_r$.
The time average growth rate is

$$\bar{g} = \mu_r + \ell\,\mu_e - \ell^2\,\frac{\sigma_s^2}{2}$$

Maximising $\bar{g}$ with respect to ℓ gives us the

Optimal leverage:

$$\ell_{opt} = \frac{\mu_e}{\sigma_s^2}$$

Comparison to standard models

Portfolio choice is usually taught through

- Markowitz portfolio theory:
  Choose a mix between the tangency portfolio and the risk-free asset.
- Sharpe ratio:
  Choose the portfolio with the highest Sharpe ratio: $\frac{\mu_s - \mu_r}{\sigma_s}$

Standard theory recognises the trade-off between risk and return but offers no guidance on how to do the trade-off (optimal choice depends on risk preferences).

Ergodicity Economics does!

Thank you.